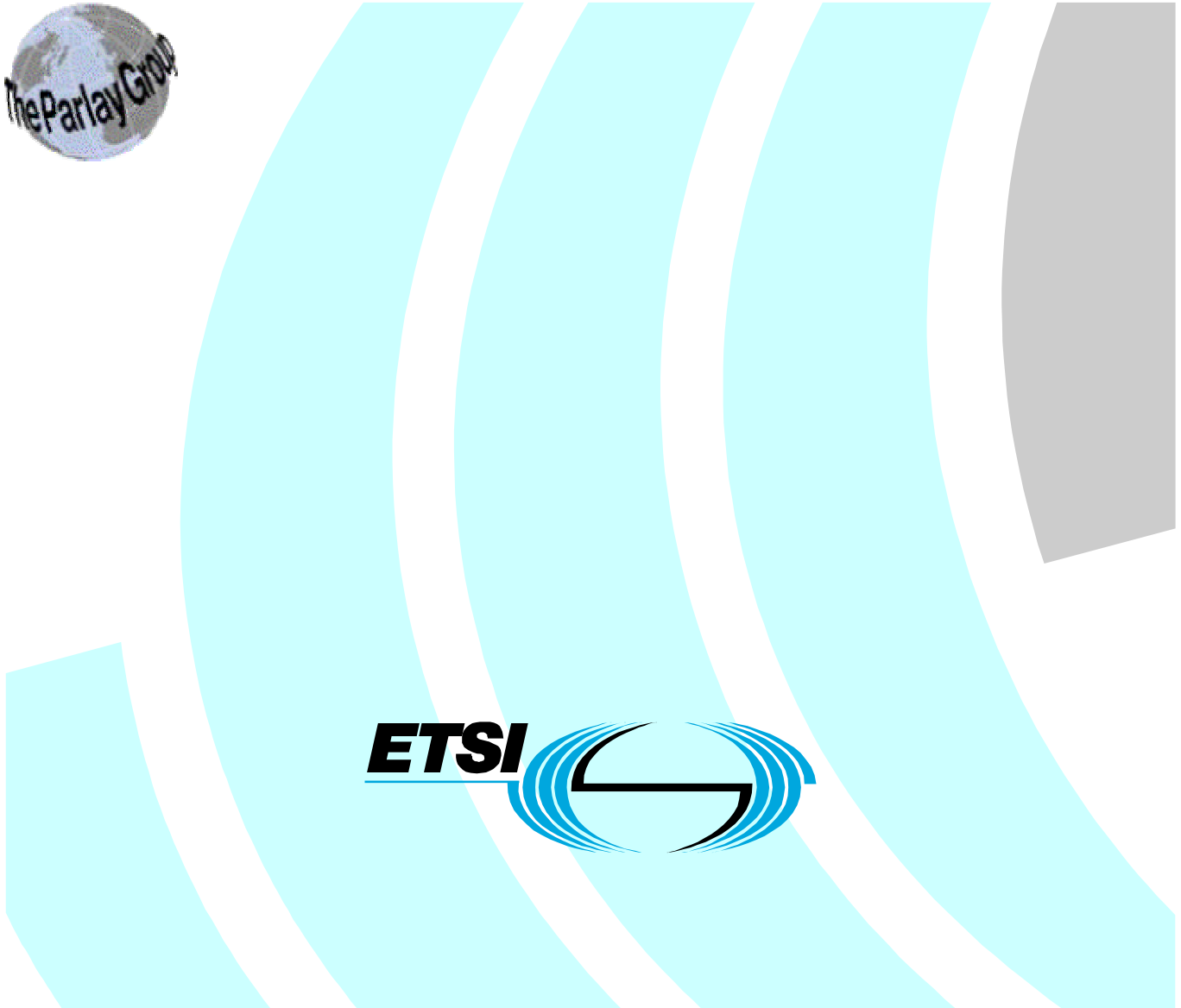
ETSI ES 202 915-4-1 V1.1.1 (2003-01)

ETSI Standard

Open Service Access (OSA);
Application Programming Interface (API);
Part 4: Call Control;
Sub-part 1: Call Control Common Definitions



Reference
DES/SPAN-120091-4-1

Keywords
API, IDL, OSA, UML

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - NAF 742 C
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Contents

Intellectual Property Rights 5
Foreword 5
1 Scope 6
2 References 6
3 Definitions and abbreviations 6
3.1 Definitions 6
3.2 Abbreviations 6
4 Call Control SCF 6
4.1 Call Model Description 7
4.2 Structure of Call Control SCF Documentation 7
5 The Service Interface Specifications 8
5.1 Interface Specification Format 8
5.1.1 Interface Class 8
5.1.2 Method descriptions 8
5.1.3 Parameter descriptions 8
5.1.4 State Model 8
5.2 Base Interface 8
5.2.1 Interface Class IpInterface 8
5.3 Service Interfaces 9
5.3.1 Overview 9
5.4 Generic Service Interface 9
5.4.1 Interface Class IpService 9
5.4.1.1 Method setCallback() 9
5.4.1.2 Method setCallbackWithSessionID() 9
6 Common Call Control Data Types 10
6.1 TpCallAlertingMechanism 10
6.2 TpCallBearerService 10
6.3 TpCallChargePlan 11
6.4 TpCallPartyToChargeAdditionalInfo 11
6.5 TpCallPartyToChargeType 11
6.6 TpCallChargeOrder 12
6.7 TpCallChargeOrderCategory 12
6.8 TpCallEndedReport 12
6.9 TpCallError 12
6.10 TpCallAdditionalErrorInfo 13
6.11 TpCallErrorType 13
6.12 TpCallInfoReport 13
6.13 TpCallInfoType 14
6.14 TpCallLoadControlMechanism 14
6.15 TpCallLoadControlIntervalRate 14
6.16 TpCallLoadControlMechanismType 14
6.17 TpCallMonitorMode 14
6.18 TpCallNetworkAccessType 15
6.19 TpCallPartyCategory 15
6.20 TpCallServiceCode 15
6.21 TpCallServiceCodeSet 15
6.22 TpCallServiceCodeType 16
6.23 TpCallSuperviseReport 16
6.24 TpCallSuperviseTreatment 16
6.25 TpCallTeleService 17
6.26 TpCallTreatment 17
6.27 TpCallTreatmentType 17
6.28 TpCallAdditionalTreatmentInfo 18

6.29 TpMediaType 18

Annex A (normative): OMG IDL Description of Common Call Control Data Types 19

Annex B (informative): W3C WSDL Description of Common Call Control Data Types 20

Annex C (informative): Java API Description of the Call Control SCFs 21

Annex D (informative): Contents of 3GPP OSA Rel-5 Call Control 22

Annex E (informative): Record of changes 23

E.1 Interfaces 23
E.1.1 New 23
E.1.2 Deprecated 23
E.1.3 Removed 23

E.2 Methods 23
E.2.1 New 23
E.2.2 Deprecated 23
E.2.3 Modified 24
E.2.4 Removed 24

E.3 Data Definitions 24
E.3.1 New 24
E.3.2 Modified 24
E.3.3 Removed 24

E.4 Service Properties 24
E.4.1 New 24
E.4.2 Deprecated 25
E.4.3 Modified 25
E.4.4 Removed 25

E.5 Exceptions 25
E.5.1 New 25
E.5.2 Modified 25
E.5.3 Removed 25

E.6 Others 25

History 26

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Foreword

This ETSI Standard (ES) has been produced by ETSI Technical Committee Services and Protocols for Advanced Networks (SPAN).

The present document is part 4, sub-part 1 of a multi-part deliverable covering Open Service Access (OSA); Application Programming Interface (API), as identified below. The API specification (ES 202 915) is structured in the following parts:

Part 1: "Overview";

Part 2: "Common Data Definitions";

Part 3: "Framework";

Part 4: "Call Control";

Sub-part 1: "Call Control Common Definitions";

Sub-part 2: "Generic Call Control SCF";

Sub-part 3: "Multi-Party Call Control SCF";

Sub-part 4: "Multi-Media Call Control SCF";

Sub-part 5: "Conference Call Control SCF";

Part 5: "User Interaction SCF";

Part 6: "Mobility SCF";

Part 7: "Terminal Capabilities SCF";

Part 8: "Data Session Control SCF";

Part 9: "Generic Messaging SCF";

Part 10: "Connectivity Manager SCF";

Part 11: "Account Management SCF";

Part 12: "Charging SCF";

Part 13: "Policy Management SCF";

Part 14: "Presence and Availability Management SCF".

The present document has been defined jointly between ETSI, The Parlay Group (http://www.parlay.org) and the 3GPP, in co-operation with a number of JAIN™ Community (http://www.java.sun.com/products/jain) member companies.

The present document forms part of the Parlay 4.0 set of specifications.

The present document is equivalent to 3GPP TS 29.198-4-1 V5.1.0 (Release 5).

1 Scope

The present document is part 4, sub-part 1 of the Stage 3 specification for an Application Programming Interface (API) for Open Service Access (OSA).

The OSA specifications define an architecture that enables application developers to make use of network functionality through an open standardised interface, i.e. the OSA APIs.

The present document specifies the common definitions used by the Call Control Service Capability Features (SCF).

2 References

The references listed in clause 2 of ES 202 915-1 contain provisions which, through reference in this text, constitute provisions of the present document.

ETSI ES 202 915-1: "Open Service Access (OSA); Application Programming Interface (API); Part 1: Overview".

ETSI ES 202 915-2: "Open Service Access (OSA); Application Programming Interface (API); Part 2: Common Data Definitions".

ETSI ES 202 915-5: "Open Service Access (OSA); Application Programming Interface (API); Part 5: User Interaction SCF".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in ES 202 915-1 apply.

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in ES 202 915-1 apply.

4 Call Control SCF

Four flavours of call control APIs are specified in ES 202 915, Parlay 4. Three of these have been included in 3GPP Release 5. These are the Generic Call Control (GCC), the Multi-Party Call Control (MPCC) and the Multi-Media Call Control (MMCC). The generic call control is the same API as was already present in the previous specification for 3GPP Release 99 (TS 129 198 V3.4.0). Multi-Party Call Control was introduced in the Release 4 specifications, and Multi-Media Call Control is introduced in Release 5. All four have been included in ETSI ES 201 915, Parlay 3.

The joint work between 3GPP CN5, ETSI SPAN and the Parlay Call Control Working group with collaboration from JAIN has been focussed on the Multi-party and Multi-Media call control APIs. A number of improvements on call control functionality have been made and are reflected in these APIs. For this it was necessary to break the inheritance that previously existed between Generic and Multi-party call control.

The joint call control group has furthermore decided that the multi-party call control is to be considered as the future base call control family and the technical work will not be continued on Generic Call control. Errors or technical flaws will of course be corrected.

4.1 Call Model Description

The call model used for the Call Control SCFs has the following objects.

- A call object. A call is a relation between a number of parties. The call object relates to the entire call view from the application. E.g. the entire call will be released when a release is called on the call. Note that different applications can have different views on the same physical call, e.g. one application for the originating side and another application for the terminating side. The applications will not be aware of each other, all 'communication' between the applications will be by means of network signalling. The API currently does not specify any feature interaction mechanisms.

- A call leg object. The leg object represents a logical association between a call and an address. The relationship includes at least the signalling relation with the party. The relation with the address is only made when the leg is routed. Before that the leg object is IDLE and not yet associated with the address.

- An address. The address logically represents a party in the call.

- A terminal. A terminal is the end-point of the signalling and/or media for a party. This object type is currently not addressed.

The call object is used to establish a relation between a number of parties by creating a leg for each party within the call.

Associated with the signalling relationship represented by the call leg, there may also be a bearer connection (e.g. in the traditional voice only networks) or a number (zero or more) of media channels (in multi-media networks).

A leg can be attached to the call or detached from the call. When the leg is attached, this means that media or bearer channels related to the legs are connected to the media or bearer channels of the other legs that are attached to the same call. I.e. only legs that are attached can 'speak' to each other. A leg can have a number of states, depending on the signalling received from or sent to the party associated with the leg. Usually there is a limit to the number of legs that are in being routed (i.e. the connection is being established) or connected to the call (i.e. the connection is established). Also, there usually is a limit to the number of legs that can be simultaneously attached to the same call.

Some networks distinguish between controlling and passive legs. By definition the call will be released when the controlling leg is released. All other legs are called passive legs. There can be at most one controlling leg per call. However, there is currently no way the application can influence whether a Leg is controlling or not.

There are two ways for an application to get the control of a call. The application can request to be notified of calls that meet certain criteria. When a call occurs in the network that meets these criteria, the application is notified and can control the call. Some legs will already be associated with the call in this case. Another way is to create a new call from the application.

4.2 Structure of Call Control SCF Documentation

Each of the Call Control SCFs is specified under the following headings:

- The Sequence diagrams give the reader a practical idea of how each of the SCF is implemented.

- The Class relationships clause shows how each of the interfaces applicable to the SCF, relate to one another.

- The Interface specification clause describes in detail each of the interfaces shown within the Class diagram part.

- The State Transition Diagrams (STD) show transition between states in the SCF. The states and transitions are well-defined; either methods specified in the Interface specification or events occurring in the underlying networks cause state transitions.

- The Data definitions clause show a detailed expansion of each of the data types associated with the methods within the classes. Note that some data types are used in other methods and classes and are therefore defined within the Common Data types part ES 202 915-2.

An implementation of one of the call control APIs which supports or implements a method described in one of the sub-parts of ES 202 915-4, shall support or implement the functionality described for that method, for at least one valid set of values for the parameters of that method. Where a method is not supported by an implementation of a Service interface, the exception P_METHOD_NOT_SUPPORTED shall be returned to any call of that method.

5 The Service Interface Specifications

5.1 Interface Specification Format

This clause defines the interfaces, methods and parameters that form a part of the API specification. The Unified Modelling Language (UML) is used to specify the interface classes. The general format of an interface specification is described below.

5.1.1 Interface Class

This shows a UML interface class description of the methods supported by that interface, and the relevant parameters and types. The Service and Framework interfaces for enterprise-based client applications are denoted by classes with name `Ip<name>`. The callback interfaces to the applications are denoted by classes with name `IpApp<name>`. For the interfaces between a Service and the Framework, the Service interfaces are typically denoted by classes with name IpSvc<name>, while the Framework interfaces are denoted by classes with name IpFw<name>

5.1.2 Method descriptions

Each method (API method "call") is described. Both synchronous and asynchronous methods are used in the API. Asynchronous methods are identified by a '`Req`' suffix for a method request, and, if applicable, are served by asynchronous methods identified by either a '`Res`' or '`Err`' suffix for method results and errors, respectively. To handle responses and reports, the application or service developer must implement the relevant `IpApp<name> or IpSvc<name>` interfaces to provide the callback mechanism.

5.1.3 Parameter descriptions

Each method parameter and its possible values are described. Parameters described as 'in' represent those that must have a value when the method is called. Those described as 'out' are those that contain the return result of the method when the method returns.

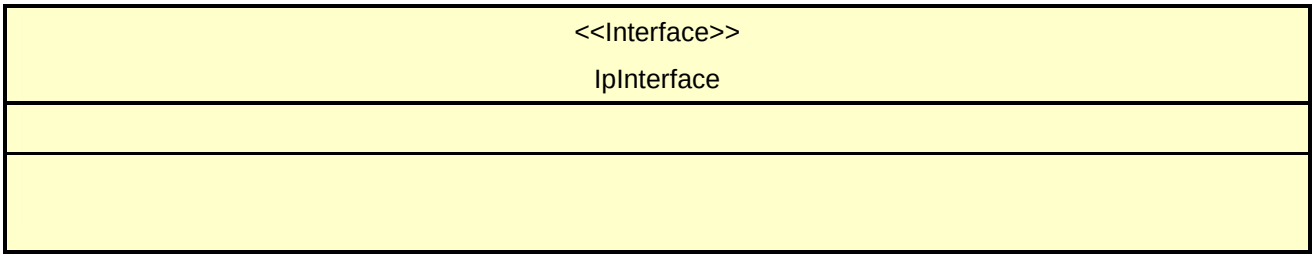
5.1.4 State Model

If relevant, a state model is shown to illustrate the states of the objects that implement the described interface.

5.2 Base Interface

5.2.1 Interface Class IpInterface

All application, framework and service interfaces inherit from the following interface. This API Base Interface does not provide any additional methods.



5.3 Service Interfaces

5.3.1 Overview

The Service Interfaces provide the interfaces into the capabilities of the underlying network - such as call control, user interaction, messaging, mobility and connectivity management.

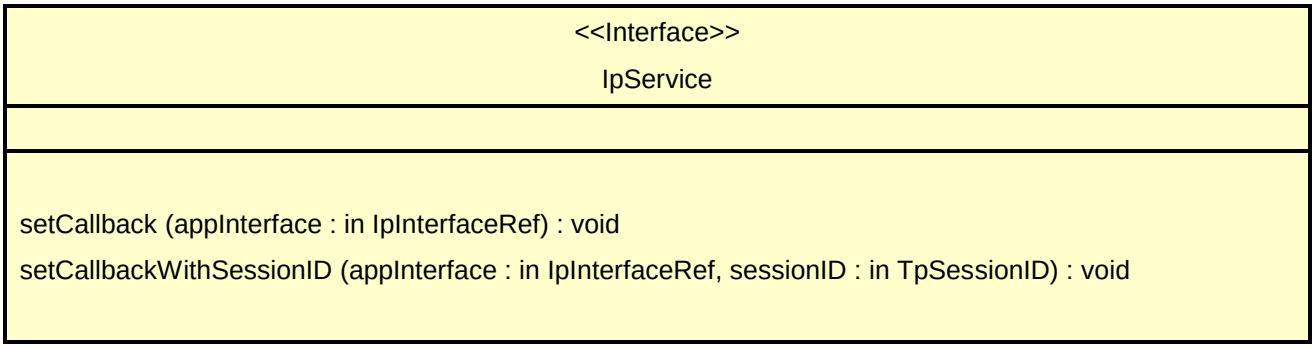
The interfaces that are implemented by the services are denoted as 'Service Interface'. The corresponding interfaces that must be implemented by the application (e.g. for API callbacks) are denoted as 'Application Interface'.

5.4 Generic Service Interface

5.4.1 Interface Class IpService

Inherits from: IpInterface

All service interfaces inherit from the following interface.



5.4.1.1 Method setCallback()

This method specifies the reference address of the callback interface that a service uses to invoke methods on the application. It is not allowed to invoke this method on an interface that uses SessionIDs.

Parameters

`appInterface : in IpInterfaceRef`

Specifies a reference to the application interface, which is used for callbacks.

Raises

`TpCommonExceptions, P_INVALID_INTERFACE_TYPE`

5.4.1.2 Method setCallbackWithSessionID()

This method specifies the reference address of the application's callback interface that a service uses for interactions associated with a specific session ID: e.g. a specific call, or call leg. It is not allowed to invoke this method on an interface that does not use SessionIDs.

Parameters

`appInterface : in IpInterfaceRef`

Specifies a reference to the application interface, which is used for callbacks.

sessionID : in TpSessionID

Specifies the session for which the service can invoke the application's callback interface.

Raises

TpCommonExceptions, P_INVALID_SESSION_ID, P_INVALID_INTERFACE_TYPE

6 Common Call Control Data Types

The following data types referenced in this clause are defined in ES 202 915-5:

`TpUIInfo`

All other data types referenced in the present document but not defined in this clause are common data definitions which may be found in ES 202 915-2.

6.1 TpCallAlertingMechanism

This data type is identical to a `TpInt32`, and defines the mechanism that will be used to alert a call party. The values of this data type are operator specific.

6.2 TpCallBearerService

This data type defines the type of call application-related specific information (ITU-T Recommendation Q.931: Information Transfer Capability, and TS 122 002)

Name	Value	Description
P_CALL_BEARER_SERVICE_UNKNOWN	0	Bearer capability information unknown at this time
P_CALL_BEARER_SERVICE_SPEECH	1	Speech
P_CALL_BEARER_SERVICE_DIGITALUNRESTRICTED	2	Unrestricted digital information
P_CALL_BEARER_SERVICE_DIGITALRESTRICTED	3	Restricted digital information
P_CALL_BEARER_SERVICE_AUDIO	4	3,1 kHz audio
P_CALL_BEARER_SERVICE_DIGITALUNRESTRICTEDTONES	5	Unrestricted digital information with tones/announcements
P_CALL_BEARER_SERVICE_VIDEO	6	Video

6.3 TpCallChargePlan

Defines the `Sequence of Data Elements` that specify the charge plan for the call.

Sequence Element Name	Sequence Element Type	Description
ChargeOrderType	TpCallChargeOrderCategory	Charge order
TransparentCharge	TpOctetSet	Operator specific charge plan specification, e.g. charging table name / charging table entry. The associated charge plan data will be send transparently to the charging records. Only applicable when transparent charging is selected.
ChargePlan	TpInt32	Pre-defined charge plan. Example of the charge plan set from which the application can choose could be : (0 = normal user, 1 = silver card user, 2 = gold card user). Only applicable when transparent charging is selected.
AdditionalInfo	TpOctetSet	Descriptive string which is sent to the billing system without prior evaluation. Could be included in the ticket.
PartyToCharge	TpCallPartyToChargeType	Identifies the entity or party to be charged for the call or call leg.
PartyToChargeAdditionalInfo	TpCallPartyToChargeAdditionalInfo	Contains additional information regarding the charged party.

6.4 TpCallPartyToChargeAdditionalInfo

Defines the Tagged Choice of Data Elements that identifies the entity or party to be charged.

	Tag Element Type	
	TpCallPartyToChargeType	

Tag Element Value	Choice Element Type	Choice Element Name
P_CALL_PARTY_ORIGINATING	NULL	Undefined
P_CALL_PARTY_DESTINATION	NULL	Undefined
P_CALL_PARTY_SPECIAL	TpAddress	CallPartySpecial

6.5 TpCallPartyToChargeType

Defines the type of call party to charge

Name	Value	Description
P_CALL_PARTY_ORIGINATING	0	Calling party, i.e. party that initiated the call. For application initiated calls this indicates the first party of the call
P_CALL_PARTY_DESTINATION	1	Called party
P_CALL_PARTY_SPECIAL	2	An address identifying e.g. a third party, a service provider

6.6 TpCallChargeOrder

Defines the `Tagged Choice of Data Elements` that specify the charge plan for the call.

	Tag Element Type	
	TpCallChargeOrderCategory	

Tag Element Value	Choice Element Type	Choice Element Name
P_CALL_CHARGE_TRANSPARENT	TpOctetSet	TransparentCharge
P_CALL_CHARGE_PREDEFINED_SET	TpInt32	ChargePlan

6.7 TpCallChargeOrderCategory

Defines the type of charging to be applied

Name	Value	Description
P_CALL_CHARGE_TRANSPARENT	0	Operator specific charge plan specification, e.g. charging table name / charging table entry. The associated charge plan data will be send transparently to the charging records
P_CALL_CHARGE_PREDEFINED_SET	1	Pre-defined charge plan. Example of the charge plan set from which the application can choose could be : (0 = normal user, 1 = silver card user, 2 = gold card user).

6.8 TpCallEndedReport

Defines the `Sequence of Data Elements` that specify the reason for the call ending.

Sequence Element Name	Sequence Element Type	Description
CallLegSessionID	TpSessionID	The leg that initiated the release of the call. If the call release was not initiated by the leg, then this value is set to -1.
Cause	TpReleaseCause	The cause of the call ending.

6.9 TpCallError

Defines the `Sequence of Data Elements` that specify the additional information relating to a call error.

Sequence Element Name	Sequence Element Type
ErrorTime	TpDateAndTime
ErrorType	TpCallErrorType
AdditionalErrorInfo	TpCallAdditionalErrorInfo

6.10 TpCallAdditionalErrorInfo

Defines the `Tagged Choice of Data Elements` that specify additional call error and call error specific information. This is also used to specify call leg errors and information errors.

	Tag Element Type	
	TpCallErrorType	

Tag Element Value	Choice Element Type	Choice Element Name
P_CALL_ERROR_UNDEFINED	NULL	Undefined
P_CALL_ERROR_INVALID_ADDRESS	TpAddressError	CallErrorInvalidAddress
P_CALL_ERROR_INVALID_STATE	NULL	Undefined
P_CALL_ERROR_RESOURCE_UNAVAILABLE	NULL	Undefined

6.11 TpCallErrorType

Defines a specific call error.

Name	Value	Description
P_CALL_ERROR_UNDEFINED	0	Undefined; the method failed or was refused, but no specific reason can be given.
P_CALL_ERROR_INVALID_ADDRESS	1	The operation failed because an invalid address was given
P_CALL_ERROR_INVALID_STATE	2	The call was not in a valid state for the requested operation
P_CALL_ERROR_RESOURCE_UNAVAILABLE	3	There are not enough resources to complete the request successfully

6.12 TpCallInfoReport

Defines the `Sequence of Data Elements` that specify the call information requested. Information that was not requested is invalid.

Sequence Element Name	Sequence Element Type	Description
CallInfoType	TpCallInfoType	The type of call report.
CallInitiationStartTime	TpDateAndTime	The time and date when the call, or follow-on call, was started.
CallConnectedToResourceTime	TpDateAndTime	The date and time when the call was connected to the resource. This data element is only valid when information on user interaction is reported.
CallConnectedToDestinationTime	TpDateAndTime	The date and time when the call was connected to the destination (i.e., when the destination answered the call). If the destination did not answer, the time is set to an empty string. This data element is invalid when information on user interaction is reported with an intermediate report.
CallEndTime	TpDateAndTime	The date and time when the call or follow-on call or user interaction was terminated.
Cause	TpReleaseCause	The cause of the termination.

A callInfoReport will be generated at the end of user interaction and at the end of the connection with the associated address. This means that either the destination related information is present or the resource related information, but not both.

6.13 TpCallInfoType

Defines the type of call information requested and reported. The values may be combined by a logical 'OR' function.

Name	Value	Description
P_CALL_INFO_UNDEFINED	00h	Undefined
P_CALL_INFO_TIMES	01h	Relevant call times
P_CALL_INFO_RELEASE_CAUSE	02h	Call release cause

6.14 TpCallLoadControlMechanism

Defines the Tagged Choice of Data Elements that specify the applied mechanism and associated parameters.

	Tag Element Type	
	TpCallLoadControlMechanismType	

Tag Element Value	Choice Element Type	Choice Element Name
P_CALL_LOAD_CONTROL_PER_INTERVAL	TpCallLoadControlIntervalRate	CallLoadControlPerInterval

6.15 TpCallLoadControlIntervalRate

Defines the call admission rate of the call load control mechanism used. This data type indicates the interval (in milliseconds) between calls that are admitted.

Name	Value	Description
P_CALL_LOAD_CONTROL_ADMIT_NO_CALLS	0	Infinite interval (do not admit any calls)
	1 to 60 000	Duration in milliseconds

6.16 TpCallLoadControlMechanismType

Defines the type of call load control mechanism to use.

Name	Value	Description
P_CALL_LOAD_CONTROL_PER_INTERVAL	0	Admit one call per interval

6.17 TpCallMonitorMode

Defines the mode that the call will monitor for events, or the mode that the call is in following a detected event.

Name	Value	Description
P_CALL_MONITOR_MODE_INTERRUPT	0	The call event is intercepted by the call control service and call processing is interrupted. The application is notified of the event and call processing resumes following an appropriate API call or network event (such as a call release)
P_CALL_MONITOR_MODE_NOTIFY	1	The call event is detected by the call control service but not intercepted. The application is notified of the event and call processing continues
P_CALL_MONITOR_MODE_DO_NOT_MONITOR	2	Do not monitor for the event

6.18 TpCallNetworkAccessType

This data defines the bearer capabilities associated with the call. (TS 124 002) This information is network operator specific and may not always be available because there is no standard protocol to retrieve the information.

Name	Value	Description
P_CALL_NETWORK_ACCESS_TYPE_UNKNOWN	0	Network type information unknown at this time
P_CALL_NETWORK_ACCESS_TYPE_POT	1	POTS
P_CALL_NETWORK_ACCESS_TYPE_ISDN	2	ISDN
P_CALL_NETWORK_ACCESS_TYPE_DIALUPINTERNET	3	Dial-up Internet
P_CALL_NETWORK_ACCESS_TYPE_XDSL	4	xDSL
P_CALL_NETWORK_ACCESS_TYPE_WIRELESS	5	Wireless

6.19 TpCallPartyCategory

This data type defines the category of a calling party. (ITU-T Recommendation Q.763: Calling Party Category / Called Party Category)

Name	Value	Description
P_CALL_PARTY_CATEGORY_UNKNOWN	0	calling party's category unknown at this time
P_CALL_PARTY_CATEGORY_OPERATOR_F	1	operator, language French
P_CALL_PARTY_CATEGORY_OPERATOR_E	2	operator, language English
P_CALL_PARTY_CATEGORY_OPERATOR_G	3	operator, language German
P_CALL_PARTY_CATEGORY_OPERATOR_R	4	operator, language Russian
P_CALL_PARTY_CATEGORY_OPERATOR_S	5	operator, language Spanish
P_CALL_PARTY_CATEGORY_ORDINARY_SUB	6	ordinary calling subscriber
P_CALL_PARTY_CATEGORY_PRIORITY_SUB	7	calling subscriber with priority
P_CALL_PARTY_CATEGORY_DATA_CALL	8	data call (voice band data)
P_CALL_PARTY_CATEGORY_TEST_CALL	9	test call
P_CALL_PARTY_CATEGORY_PAYPHONE	10	payphone

6.20 TpCallServiceCode

Defines the `Sequence of Data Elements` that specify the service code and type of service code received during a call. The service code type defines how the value string should be interpreted.

Sequence Element Name	Sequence Element Type
CallServiceCodeType	TpCallServiceCodeType
ServiceCodeValue	TpString

6.21 TpCallServiceCodeSet

Defines a Numbered Set of Data Elements of TpCallServiceCode.

6.22 TpCallServiceCodeType

Defines the different types of service codes that can be received during the call.

Name	Value	Description
P_CALL_SERVICE_CODE_UNDEFINED	0	The type of service code is unknown. The corresponding string is operator specific.
P_CALL_SERVICE_CODE_DIGITS	1	The user entered a digit sequence during the call. The corresponding string is an ASCII representation of the received digits.
P_CALL_SERVICE_CODE_FACILITY	2	A facility information element is received. The corresponding string contains the facility information element as defined in ITU-T Recommendation Q.932.
P_CALL_SERVICE_CODE_U2U	3	A user-to-user message was received. The associated string contains the content of the user-to-user information element.
P_CALL_SERVICE_CODE_HOOKFLASH	4	The user performed a hookflash, optionally followed by some digits. The corresponding string is an ASCII representation of the entered digits.
P_CALL_SERVICE_CODE_RECALL	5	The user pressed the register recall button, optionally followed by some digits. The corresponding string is an ASCII representation of the entered digits.

6.23 TpCallSuperviseReport

Defines the responses from the call control service for calls that are supervised. The values may be combined by a logical 'OR' function.

Name	Value	Description
P_CALL_SUPERVISE_TIMEOUT	01h	The call supervision timer has expired.
P_CALL_SUPERVISE_CALL_ENDED	02h	The call has ended, either due to timer expiry or call party release. In case the called party disconnects but a follow-on call can still be made also this indication is used.
P_CALL_SUPERVISE_TONE_APPLIED	04h	A warning tone has been applied. This is only sent in combination with P_CALL_SUPERVISE_TIMEOUT
P_CALL_SUPERVISE_UI_FINISHED	08h	The user interaction has finished.

6.24 TpCallSuperviseTreatment

Defines the treatment of the call by the call control service when the call supervision timer expires. The values may be combined by a logical 'OR' function.

Name	Value	Description
P_CALL_SUPERVISE_RELEASE	01h	Release the call when the call supervision timer expires
P_CALL_SUPERVISE_RESPOND	02h	Notify the application when the call supervision timer expires
P_CALL_SUPERVISE_APPLY_TONE	04h	Send a warning tone to the originating party when the call supervision timer expires. If call release is requested, then the call will be released following the tone after an administered time period

6.25 TpCallTeleService

This data type defines the tele-service associated with the call. (ITU-T Recommendation Q.763: User Teleservice Information, ITU-T Recommendation Q.931: High Layer Compatibility Information, and TS 122 003)

Name	Value	Description
P_CALL_TELE_SERVICE_UNKNOWN	0	Teleservice information unknown at this time
P_CALL_TELE_SERVICE_TELEPHONY	1	Telephony
P_CALL_TELE_SERVICE_FAX_2_3	2	Facsimile Group 2/3
P_CALL_TELE_SERVICE_FAX_4_I	3	Facsimile Group 4, Class I
P_CALL_TELE_SERVICE_FAX_4_II_III	4	Facsimile Group 4, Classes II and III
P_CALL_TELE_SERVICE_VIDEOTEX_SYN	5	Syntax based Videotex
P_CALL_TELE_SERVICE_VIDEOTEX_INT	6	International Videotex interworking via gateways or interworking units
P_CALL_TELE_SERVICE_TELEX	7	Telex service
P_CALL_TELE_SERVICE_MHS	8	Message Handling Systems
P_CALL_TELE_SERVICE_OSI	9	OSI application
P_CALL_TELE_SERVICE_FTAM	10	FTAM application
P_CALL_TELE_SERVICE_VIDEO	11	Videotelephony
P_CALL_TELE_SERVICE_VIDEO_CONF	12	Videoconferencing
P_CALL_TELE_SERVICE_AUDIOGRAPH_CONF	13	Audiographic conferencing
P_CALL_TELE_SERVICE_MULTIMEDIA	14	Multimedia services
P_CALL_TELE_SERVICE_CS_INI_H221	15	Capability set of initial channel of ITU-T Recommendation H.221
P_CALL_TELE_SERVICE_CS_SUB_H221	16	Capability set of subsequent channel of ITU-T Recommendation H.221
P_CALL_TELE_SERVICE_CS_INI_CALL	17	Capability set of initial channel associated with an active 3,1 kHz audio or speech call.
P_CALL_TELE_SERVICE_DATATRAFFIC	18	Data traffic.
P_CALL_TELE_SERVICE_EMERGENCY_CALLS	19	Emergency Calls
P_CALL_TELE_SERVICE_SMS_MT_PP	20	Short message MT/PP
P_CALL_TELE_SERVICE_SMS_MO_PP	21	Short message MO/PP
P_CALL_TELE_SERVICE_CELL_BROADCAST	22	Cell Broadcast Service
P_CALL_TELE_SERVICE_ALT_SPEECH_FAX_3	23	Alternate speech and facsimile group 3
P_CALL_TELE_SERVICE_AUTOMATIC_FAX_3	24	Automatic Facsimile group 3
P_CALL_TELE_SERVICE_VOICE_GROUP_CALL	25	Voice Group Call Service
P_CALL_TELE_SERVICE_VOICE_BROADCAST	26	Voice Broadcast Service

6.26 TpCallTreatment

Defines the `Sequence of Data Elements` that specify the treatment for calls that will be handled only by the network (for example, call which are not admitted by the call load control mechanism).

Sequence Element Name	Sequence Element Type
CallTreatmentType	TpCallTreatmentType
ReleaseCause	TpReleaseCause
AdditionalTreatmentInfo	TpCallAdditionalTreatmentInfo

6.27 TpCallTreatmentType

Defines the treatment for calls that will be handled only by the network.

Name	Value	Description
P_CALL_TREATMENT_DEFAULT	0	Default treatment
P_CALL_TREATMENT_RELEASE	1	Release the call
P_CALL_TREATMENT_SIAR	2	Send information to the user, and release the call (Send Info & Release)

6.28 TpCallAdditionalTreatmentInfo

Defines the `Tagged Choice of Data Elements` that specify the information to be sent to a call party.

	Tag Element Type	
	TpCallTreatmentType	

Tag Element Value	Choice Element Type	Choice Element Name
P_CALL_TREATMENT_DEFAULT	NULL	Undefined
P_CALL_TREATMENT_RELEASE	NULL	Undefined
P_CALL_TREATMENT_SIAR	TpUIInfo	InformationToSend

6.29 TpMediaType

Defines the media type of a media stream. The values may be combined by a logical 'OR' function.

Name	Value	Description
P_AUDIO	1	Audio stream
P_VIDEO	2	Video stream
P_DATA	4	Data stream (e.g. T.120)

Annex A (normative): OMG IDL Description of Common Call Control Data Types

The OMG IDL representation of this interface specification is contained in a text file (common_cc_data.idl contained in archive es_2029150401v010101p0.ZIP) which accompanies the present document.

Annex B (informative): W3C WSDL Description of Common Call Control Data Types

The W3C WSDL representation of this interface specification is contained in a text file (common_cc_data.wsdl contained in archive es_2029150401v010101p0.ZIP) which accompanies the present document.

Annex C (informative): Java API Description of the Call Control SCFs

The Java API representation of this interface specification can be obtained from the following URL:

- Java Call Control (http://jcp.org/jsr/detail/21.jsp)

Each JSR webpage contains a table identifying the relationships between the different versions of the Parlay, ETSI/OSA, 3GPP/OSA and JAIN SPA specifications. In addition, each JAIN SPA specification version indicates to which Parlay, ETSI/OSA and 3GPP/OSA specification versions it corresponds to.

Annex D (informative): Contents of 3GPP OSA Rel-5 Call Control

All items in the present document are relevant for TS 129 198-4-1 V5 (Release 5).

Annex E (informative): Record of changes

The following is a list of the changes made to the present document for each release. The list contains the names of all changed, deprecated, added or removed items in the specifications and not the actual changes. Any type of change information that is important to the reader is put in the final clause of this annex.

Changes are specified as changes to the prior major release, but every minor release will have its own part of the table allowing the reader to know when the actual change was made.

E.1 Interfaces

E.1.1 New

Identifier	Comments
Interfaces added in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.1.2 Deprecated

Identifier	Comments
Interfaces deprecated in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.1.3 Removed

Identifier	Comments
Interfaces removed in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.2 Methods

E.2.1 New

Identifier	Comments
Methods added in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.2.2 Deprecated

Identifier	Comments
Methods deprecated in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.2.3 Modified

Identifier	Comments
Methods modified in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.2.4 Removed

Identifier	Comments
Methods removed in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.3 Data Definitions

E.3.1 New

Identifier	Comments
Data Definitions added in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.3.2 Modified

Identifier	Comments
Data Definitions modified in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.3.3 Removed

Identifier	Comments
Data Definitions removed in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.4 Service Properties

E.4.1 New

Identifier	Comments
Service Properties added in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.4.2 Deprecated

Identifier	Comments
Service Properties deprecated in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.4.3 Modified

Identifier	Comments
Service Properties modified in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.4.4 Removed

Identifier	Comments
Service Properties removed in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.5 Exceptions

E.5.1 New

Identifier	Comments
Exceptions added in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.5.2 Modified

Identifier	Comments
Exceptions modified in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.5.3 Removed

Identifier	Comments
Exceptions removed in ES 202 915-4-1 version 1.1.1 (Parlay 4.0)	

E.6 Others

History

Document history			
V1.1.1	November 2002	Membership Approval Procedure	MV 20030117: 2002-11-19 to 2003-01-17
V1.1.1	January 2003	Publication	